

VOYANA

AI-Powered Collaborative Travel Ecosystem

BUSINESS REQUIREMENTS DOCUMENT

Functional and Non-Functional Requirements | User Stories and Use Case Diagrams | Interface Requirements | Wireframes and UI/UX Mockups | Acceptance Criteria

Field	Detail
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1. Functional Requirements

Requirements are grouped by service module and identified as FR-<MODULE>-<NN>. Priority uses MoSCoW: M (Must have, MVP-blocking), S (Should have), C (Could have), W (Will not have this release). Every requirement is written so that a tester can determine pass or fail without consulting the author.

Prefix	Module	Jira epic	Owner
FR-AUT	Authentication	VPM-2	Nigam Sanghvi
FR-USR	User Management	VPM-2	Nigam Sanghvi
FR-TRP	Trip Planning	VPM-4	Megha Lalwani
FR-AI	AI Services	VPM-5	Bhavika Mulani / Jagrat Kumar
FR-COL	Collaboration and Trip Workspace	VPM-8	Megha Lalwani
FR-BKG	Booking Services	VPM-6	Bhavika Sainani
FR-EXP	Expense Management	VPM-7	Jagrat Kumar
FR-PAY	Payment	VPM-7	Jagrat Kumar / Bhavika Sainani
FR-NOT	Notifications	VPM-9	Manav Vyas
FR-DOC	Document Management	VPM-9	Manav Vyas / Diya Shah
FR-MEM	Travel Memories	VPM-9	Manav Vyas / Tirth Gandhi
FR-ADM	Analytics and Administration	VPM-9 / VPM-10	Diya Shah

1.1 Authentication (FR-AUT)

ID	Requirement	Pri.	Verification
FR-AUT-01	The system shall allow a visitor to register with a full name, a unique email address and a password, rejecting any email already held by an active account.	M	Test case, negative path
FR-AUT-02	The system shall enforce a password policy of at least eight characters containing one uppercase letter, one lowercase letter, one numeral and one symbol, and shall display live strength feedback during entry.	M	Test case, boundary values
FR-AUT-03	The system shall email a six-digit verification code on registration, expiring the code after ten minutes and permitting at most five verification attempts per code.	M	Test case, timing
FR-AUT-04	The system shall refuse access to trip functionality until the account email is verified.	M	Test case
FR-AUT-05	The system shall support federated sign-in through Google and Apple OAuth 2.0, creating a Voyana account on first use and linking to an existing account when the verified email matches.	M	Integration test
FR-AUT-06	The system shall issue a JWT access token valid for 15 minutes and a refresh token valid for 7 days, and shall rotate the refresh token on each use.	M	Integration test
FR-AUT-07	The system shall allow a user to request a password reset link that expires after 30 minutes and is single use.	M	Test case

ID	Requirement	Pri.	Verification
FR-AUT-08	The system shall list active sessions with device and last-active time and allow the user to revoke any session individually or all sessions at once.	S	Test case
FR-AUT-09	The system shall lock an account for 15 minutes after five consecutive failed sign-in attempts and notify the account holder by email.	M	Test case, negative path
FR-AUT-10	The system shall apply role-based access control with the roles traveller, organizer and administrator, and shall re-verify the role at every service boundary rather than trusting the client.	M	Code review and test

1.2 User Management (FR-USR)

ID	Requirement	Pri.	Verification
FR-USR-01	The system shall allow a user to maintain a profile containing display name, profile photo, home city, preferred currency and travel preferences.	M	Test case
FR-USR-02	The system shall allow a user to record travel preferences across budget band, pace, cuisine, accessibility needs and interest tags, and shall make these available to the AI Service as generation context.	M	Integration test
FR-USR-03	The system shall allow a user to maintain a connections list of frequent travel companions and to invite from that list in one action.	S	Test case
FR-USR-04	The system shall allow a user to export all personal data held about them as a machine-readable archive within 24 hours of request.	S	Test case
FR-USR-05	The system shall allow a user to deactivate their account, after which the profile becomes inaccessible while trip records retain an anonymised reference for expense integrity.	M	Test case
FR-USR-06	The system shall prevent deactivation while the user holds an unsettled balance on any active trip, and shall state the blocking amount.	S	Test case, negative path

1.3 Trip Planning (FR-TRP)

ID	Requirement	Pri.	Verification
FR-TRP-01	The system shall allow an authenticated user to create a trip with a title, one or more destinations, a start and end date and a total budget with currency.	M	Test case
FR-TRP-02	The system shall reject a trip whose end date precedes its start date and shall warn when the duration exceeds 60 days.	M	Test case, boundary
FR-TRP-03	The system shall generate a day container for each date in the trip range and shall re-map existing activities when the range is later changed, flagging any activity that falls outside the new range.	M	Test case
FR-TRP-04	The system shall allow a member to add, edit, reorder and delete itinerary activities, each carrying a title, time, duration, location, category and estimated cost per person.	M	Test case

ID	Requirement	Pri.	Verification
FR-TRP-05	The system shall allow an organizer to invite members by email, username or shareable link, with links expiring after seven days.	M	Test case
FR-TRP-06	The system shall present a trip dashboard showing dates, members, planning progress, budget consumed and the next upcoming activity.	M	Test case
FR-TRP-07	The system shall allow a trip to be saved as a reusable template and a template to be instantiated as a new trip with dates rebased to the new start date.	S	Test case
FR-TRP-08	The system shall apply optimistic locking to itinerary edits, rejecting a stale write with a conflict response and presenting the current value alongside the attempted value. Conflict-free real-time editing is deferred to Phase 2.	M	Integration test
FR-TRP-09	The system shall maintain an itinerary change log recording the actor, the timestamp and the previous value for every change, retained for the life of the trip.	M	Test case
FR-TRP-10	The system shall allow a trip to be shared read-only through a public link that hides expense figures, member contact details and stored documents.	S	Test case, security
FR-TRP-11	The system shall allow an organizer to archive a completed trip, keeping it readable while blocking further edits.	C	Test case

1.4 AI Services (FR-AI)

ID	Requirement	Pri.	Verification
FR-AI-01	The system shall generate a day-by-day itinerary from destination, date range, budget, group size and stated preferences, returning between three and six activities per day with time, duration and estimated cost.	M	Test case, UAT
FR-AI-02	The system shall present a generated itinerary as an editable draft that a member must accept before it replaces or merges with the existing plan.	M	Test case
FR-AI-03	The system shall produce a budget plan allocating the trip budget across accommodation, transport, food, activities and contingency, and shall flag any plan whose estimated total exceeds the stated budget.	M	Test case
FR-AI-04	The system shall provide a conversational assistant that answers questions with the current trip, itinerary and budget loaded as context, and shall cite which trip elements informed each answer.	M	Test case, UAT
FR-AI-05	The system shall recommend destinations from stated preferences, season and budget, returning at least five options each with a stated reason.	S	Test case
FR-AI-06	The system shall generate a packing checklist from destination, season, duration and planned activity categories, and shall write it to the shared packing list.	S	Test case
FR-AI-07	The system shall retrieve the forecast for each itinerary day and shall propose an alternative when an outdoor activity coincides with adverse weather, requiring member confirmation before any change.	S	Integration test

ID	Requirement	Pri.	Verification
FR-AI-08	The system shall compile a travel journal from the itinerary, logged expenses and uploaded memories, returning an editable draft.	S	Test case
FR-AI-09	The system shall provide visa and entry guidance carrying an explicit disclaimer that it is not official advice and directing the user to the relevant consular authority.	S	Content review
FR-AI-10	The system shall optimise the order of activities within a single day to reduce total travel time, presenting the saving achieved and requiring confirmation before applying.	C	Test case
FR-AI-11	The system shall log every AI request and response with the trip identifier, prompt version, token count and latency, retaining the log for 30 days for evaluation.	M	Log inspection
FR-AI-12	The system shall return a clear failure message and preserve manual editing capability whenever the AI provider is unavailable or rate limited.	M	Fault injection test

1.5 Collaboration and Trip Workspace (FR-COL)

ID	Requirement	Pri.	Verification
FR-COL-01	The system shall provide each trip with a workspace containing chat, polls, tasks, a shared calendar and a shared packing list, accessible only to members of that trip.	M	Test case, security
FR-COL-02	The system shall deliver group chat messages to all connected members within two seconds and shall persist the full message history for the life of the trip.	M	Performance test
FR-COL-03	The system shall present a shared calendar that reflects itinerary activities, booking times and task due dates in a single view.	M	Test case
FR-COL-04	The system shall allow any member to create a task with a title, assignee and due date, and shall allow the assignee or an organizer to mark it complete.	M	Test case
FR-COL-05	The system shall allow a member to open a poll with two to six options and a closing time, shall accept one vote per member, shall display live tallies and shall record the outcome and the closing actor when the poll closes.	M	Test case
FR-COL-06	The system shall allow the winning option of an activity poll to be added to the itinerary in a single action.	S	Test case
FR-COL-07	The system shall maintain a shared packing list on which any member may claim an item, showing the claiming member against that item.	S	Test case
FR-COL-08	The system shall notify members when they are mentioned in chat, assigned a task or added to a poll.	M	Integration test
FR-COL-09	The system shall present a live travel updates feed for the trip. Deferred to Phase 2 pending partner webhooks.	W	Not tested this release

1.6 Booking Services (FR-BKG)

ID	Requirement	Pri.	Verification
FR-BKG-01	The system shall allow a member to search flights by origin, destination, dates and passenger count, returning results with	M	Integration test

ID	Requirement	Pri.	Verification
	carrier, times, duration, stop count and total price inclusive of taxes.		
FR-BKG-02	The system shall allow a member to search accommodation by destination, dates, guest count and price band, returning results with location, rating, cancellation terms and total price.	M	Integration test
FR-BKG-03	The system shall allow a member to arrange local transport at the destination, including airport transfers and vehicle rental.	S	Integration test
FR-BKG-04	The system shall present a single confirmation view per trip listing every flight, stay and transport reservation with its reference, status and cancellation deadline.	M	Test case
FR-BKG-05	The system shall write every confirmed booking to the trip itinerary and to the expense log as a single action, so that a booking never exists without a corresponding budget entry.	M	Integration test
FR-BKG-06	The system shall access partner inventory through an adapter interface so that a sandbox provider can be replaced by a commercial provider without changing the Booking Service.	M	Code review
FR-BKG-07	The system shall allow a booking to be cancelled or amended where partner terms permit, shall display the applicable penalty before confirmation and shall reverse the associated expense entry on success.	M	Test case
FR-BKG-08	The system shall restrict confirmation of a booking that draws on the trip budget to an organizer.	M	Test case, security
FR-BKG-09	The system shall hold a search result price for ten minutes and shall re-verify the price with the partner before confirmation, presenting any change for acceptance.	S	Integration test
FR-BKG-10	The system shall retry a failed partner call twice with exponential backoff and shall surface a clear failure state without creating a partial booking record.	M	Fault injection test

1.7 Expense Management (FR-EXP)

ID	Requirement	Pri.	Verification
FR-EXP-01	The system shall allow a member to log an expense with an amount, currency, date, category, payer and the set of members sharing it.	M	Test case
FR-EXP-02	The system shall split an expense equally by default and shall support unequal splits by fixed amount or by percentage, rejecting any split that does not reconcile to the expense total.	M	Test case, boundary
FR-EXP-03	The system shall compute each member's net position across all trip expenses and shall reduce the resulting obligations to the minimum number of transfers.	M	Automated test
FR-EXP-04	The system shall display budget against actual spend for the trip and per category, and shall raise a visible warning at 80 percent and at 100 percent of the trip budget.	M	Test case
FR-EXP-05	The system shall record a settlement when a member marks a transfer as paid, shall require confirmation from the receiving member and shall update both net positions on confirmation.	M	Test case
FR-EXP-06	The system shall accept expenses in any supported currency, shall convert to the trip currency using the rate on the expense	M	Test case

ID	Requirement	Pri.	Verification
	date and shall store both the original and converted values with the rate applied.		
FR-EXP-07	The system shall allow an expense to carry an attached receipt image of up to 5 MB.	S	Test case
FR-EXP-08	The system shall allow only the expense creator or an organizer to edit or delete an expense, and shall record every such change in an audit log.	M	Test case, security
FR-EXP-09	The system shall export the trip expense log as CSV.	C	Test case

1.8 Payment (FR-PAY)

ID	Requirement	Pri.	Verification
FR-PAY-01	The system shall initiate payment for a booking through the integrated gateway in test mode and shall confirm the booking only after the gateway reports success.	M	Integration test
FR-PAY-02	The system shall never receive, log or store raw card data; all card capture shall occur within the gateway-hosted component.	M	Code review, security
FR-PAY-03	The system shall handle gateway webhooks idempotently, so that a repeated notification produces no duplicate booking, expense or settlement.	M	Integration test
FR-PAY-04	The system shall present a payment history per user and per trip showing amount, method, status, timestamp and a downloadable receipt.	M	Test case
FR-PAY-05	The system shall allow one member to settle a balance with another in the application. Live money movement deferred to Phase 2; test mode only this release.	S	Test case, test mode
FR-PAY-06	The system shall process a refund for a cancelled booking through the gateway and shall reverse the linked expense entry on confirmation.	M	Integration test
FR-PAY-07	The system shall mark a payment as failed and preserve the underlying cart or settlement intent when the gateway declines or times out, allowing the user to retry.	M	Fault injection test

1.9 Notifications (FR-NOT)

ID	Requirement	Pri.	Verification
FR-NOT-01	The system shall maintain an in-application notification centre showing unread and read notifications for the last 90 days.	M	Test case
FR-NOT-02	The system shall deliver web push notifications for booking confirmations, booking changes, itinerary changes, poll closures, task assignments and settlement requests.	M	Integration test
FR-NOT-03	The system shall allow a user to enable or disable each notification category independently for in-application and push channels.	M	Test case
FR-NOT-04	The system shall schedule reminders for check-in windows, cancellation deadlines, document expiry and task due dates, and shall cancel a reminder automatically when its underlying event is removed.	M	Test case

ID	Requirement	Pri.	Verification
FR-NOT-05	The system shall suppress duplicate notifications for the same event and recipient within a five-minute window.	S	Test case
FR-NOT-06	The system shall fall back to in-application notification when push permission is absent or delivery fails.	M	Fault injection test

1.10 Document Management (FR-DOC)

ID	Requirement	Pri.	Verification
FR-DOC-01	The system shall allow a user to upload travel documents in PDF, JPEG or PNG format up to 10 MB per file into a per-trip vault.	M	Test case
FR-DOC-02	The system shall encrypt stored documents at rest and shall serve them only through time-limited signed URLs valid for no more than 15 minutes.	M	Security test
FR-DOC-03	The system shall allow the uploading user to grant or withhold visibility of a document to other trip members, defaulting to private.	M	Test case, security
FR-DOC-04	The system shall record an expiry date for identity documents and shall raise a reminder at 180, 90 and 30 days before expiry.	S	Test case
FR-DOC-05	The system shall record an access log for every document view and download, visible to the uploading user.	S	Test case

1.11 Travel Memories (FR-MEM)

ID	Requirement	Pri.	Verification
FR-MEM-01	The system shall allow a member to upload photographs and notes to a trip, up to 20 files per upload action.	M	Test case
FR-MEM-02	The system shall associate each uploaded item with a specific itinerary day, defaulting to the day matching the capture timestamp where available and prompting for selection otherwise.	M	Test case
FR-MEM-03	The system shall present a trip album ordered by itinerary day with the associated activity shown against each group of items.	M	Test case
FR-MEM-04	The system shall generate a caption suggestion for an uploaded photograph, which the uploader may accept or replace.	C	Test case
FR-MEM-05	The system shall restrict album visibility to trip members unless the uploader explicitly shares an item.	M	Test case, security
FR-MEM-06	The system shall allow an uploader to delete their own items, removing them from storage within 24 hours.	M	Test case

1.12 Analytics and Administration (FR-ADM)

ID	Requirement	Pri.	Verification
FR-ADM-01	The system shall allow an administrator to search, view, verify and suspend traveller and partner accounts, recording a reason against every action.	M	Test case
FR-ADM-02	The system shall present a moderation queue of reported or automatically flagged reviews and memories, allowing approval or removal with a recorded reason.	M	Test case

ID	Requirement	Pri.	Verification
FR-ADM-03	The system shall present a service health view showing the status and 95th-percentile latency of every service.	S	Test case
FR-ADM-04	The system shall write every administrative action to an immutable audit log recording actor, action, target and timestamp.	M	Log inspection
FR-ADM-05	The system shall present a personal travel spend dashboard to each user covering trips, categories and time.	S	Test case
FR-ADM-06	The system shall present organizer-level engagement analytics. Deferred to Phase 2.	W	Not tested this release
FR-ADM-07	The system shall present platform-wide usage analytics. Demonstrated with seeded data this release.	C	Demonstration

1.13 Role Permission Matrix

Permissions are enforced by the Auth Service and re-verified by every downstream service. Trip-scoped roles apply only within the trip that granted them.

Capability	Guest	Traveller	Organizer	Admin
Create a trip	No	Yes	Yes	Yes
Invite or remove trip members	No	No	Yes (own trip)	Yes
Edit the shared itinerary	No	Yes (member)	Yes	Yes
Open, close or delete a poll	No	Open only	All	All
Confirm a booking against the trip budget	No	No	Yes	No
Log an expense	No	Yes	Yes	No
Delete another member's expense	No	No	Yes	Yes
Upload to the document vault	No	Yes (own)	Yes	No
Moderate reviews and memories	No	No	No	Yes
Suspend or verify an account	No	No	No	Yes
View platform-wide analytics	No	No	No	Yes

2. Non-Functional Requirements

Non-functional requirements state measurable quality targets. Each is written with a threshold and a verification method so that it can be tested rather than asserted. Targets are calibrated to an academic deployment with a UAT panel, not to production-scale traffic.

2.1 Performance

ID	Requirement	Target	Verification
NFR-PRF-01	Application programming interface responses for read operations, measured at the gateway under a load of 50 concurrent users.	p95 under 400 ms	Load test, k6
NFR-PRF-02	Application programming interface responses for write operations under the same load.	p95 under 800 ms	Load test, k6
NFR-PRF-03	First contentful paint of the trip dashboard on a mid-range mobile device over a 4G connection.	Under 2.0 s	Lighthouse
NFR-PRF-04	Group chat message delivery from send to receipt by connected members.	Under 2 s	Instrumented test
NFR-PRF-05	AI itinerary generation for a seven-day, five-member trip, measured end to end.	Under 30 s, p95	Timed test, 20 runs
NFR-PRF-06	Settlement calculation for a trip holding 200 expenses across 10 members.	Under 1 s	Automated test
NFR-PRF-07	Search result rendering for 50 flight results.	Under 1.5 s	Instrumented test

2.2 Scalability

ID	Requirement	Target	Verification
NFR-SCL-01	Each service shall scale horizontally without change to application code or shared session state.	Stateless services	Design review, scale test
NFR-SCL-02	The platform shall sustain the demonstration load without degradation beyond the performance targets.	200 concurrent users	Load test
NFR-SCL-03	A single trip shall support members, itinerary activities and expenses at the stated volumes without functional degradation.	25 members, 300 activities, 500 expenses	Seeded data test
NFR-SCL-04	AI Service instances shall scale independently of all other services.	Independent scaling	Deployment evidence
NFR-SCL-05	Media storage shall accept new capacity without service interruption.	No downtime	Design review

2.3 Security

ID	Requirement	Target	Verification
NFR-SEC-01	All traffic between client and platform, and between services, shall be encrypted in transit.	TLS 1.2 or higher	Configuration audit
NFR-SEC-02	Passwords shall be stored using a memory-hard adaptive hash with a per-user salt.	bcrypt, cost 12 or higher	Code review
NFR-SEC-03	Personally identifying data and stored documents shall be encrypted at rest.	AES-256	Configuration audit

ID	Requirement	Target	Verification
NFR-SEC-04	Every service shall validate the JWT and re-check authorisation on every request; no service shall trust a client-supplied role claim.	Zero trust between services	Code review, penetration test
NFR-SEC-05	All input shall be validated server side against an explicit schema, and all output shall be encoded to prevent injection and cross-site scripting.	Zero critical findings	Static analysis, OWASP ZAP
NFR-SEC-06	Authentication endpoints shall be rate limited per address and per account.	10 requests per minute	Test case
NFR-SEC-07	Raw payment card data shall never traverse or persist within Voyana infrastructure.	Zero exposure	Code review, gateway audit
NFR-SEC-08	Dependencies shall carry no known critical or high severity vulnerability at release.	Zero critical or high	npm audit in CI
NFR-SEC-09	Trip data shall be inaccessible to any user who is not a member of that trip, verified by explicit negative test for every trip-scoped endpoint.	Zero leakage	Automated security suite

2.4 Availability and Reliability

ID	Requirement	Target	Verification
NFR-AVL-01	Platform availability measured across the demonstration and evaluation window.	99.5 percent	Uptime monitor
NFR-AVL-02	Failure of a non-critical service shall not prevent core planning, itinerary and expense functions.	Graceful degradation	Fault injection test
NFR-AVL-03	Database backups shall be taken automatically and shall be restorable.	Daily, restore under 4 h	Restore drill
NFR-AVL-04	Every AI-assisted screen shall offer a manual path when the AI Service is unavailable.	100 percent of AI screens	Fault injection test
NFR-AVL-05	No booking, expense or settlement shall be lost or duplicated as a result of a partner or gateway failure.	Zero data loss	Fault injection test
NFR-AVL-06	Every service shall expose a health endpoint reporting its own status and that of its dependencies.	All 12 services	Test case

2.5 Usability and Accessibility

ID	Requirement	Target	Verification
NFR-USA-01	A first-time user shall create a trip and generate an itinerary without assistance.	Under 5 minutes, 8 of 10 users	UAT observation
NFR-USA-02	All primary screens shall render correctly at the stated viewport widths without horizontal scrolling.	360, 768, 1024, 1440 px	Cross-device review
NFR-USA-03	Every destructive action shall require explicit confirmation naming the object being affected.	100 percent of destructive actions	Interface review
NFR-USA-04	The interface shall meet WCAG 2.1 level AA for contrast, keyboard navigation, focus order and form labelling.	AA, self-assessed	axe DevTools audit
NFR-USA-05	Every error message shall state what failed and what the user can do next, without exposing internal identifiers or stack traces.	100 percent of error states	Interface review

ID	Requirement	Target	Verification
NFR-USA-06	Any operation exceeding one second shall display a progress indicator.	100 percent of long operations	Interface review
NFR-USA-07	AI-generated content shall be visibly labelled as such wherever it is presented.	100 percent of AI output	Interface review

2.6 Compatibility

ID	Requirement	Target	Verification
NFR-CMP-01	The application shall function on current and immediately previous major versions of Chrome, Firefox, Safari and Edge.	Four browsers, two versions	Cross-browser test
NFR-CMP-02	The application shall function on iOS Safari 16 and later and Android Chrome 110 and later.	Two mobile platforms	Device test
NFR-CMP-03	All interfaces shall exchange JSON over REST and shall be described by an OpenAPI 3.0 document.	12 of 12 services	Contract review

2.7 Maintainability and Supportability

ID	Requirement	Target	Verification
NFR-MNT-01	Each service shall build, test and deploy independently through the continuous integration pipeline.	12 independent pipelines	Pipeline evidence
NFR-MNT-02	Automated statement coverage on service business logic.	At least 70 percent	Jest coverage report
NFR-MNT-03	Every service shall emit structured logs carrying a correlation identifier that persists across service boundaries.	100 percent of requests	Log inspection
NFR-MNT-04	Every published interface shall be versioned, and a breaking change shall require a new version path.	Semantic versioning	Contract review
NFR-MNT-05	The codebase shall pass lint and format checks in continuous integration with no warnings suppressed without a recorded reason.	Zero unexplained suppressions	Pipeline evidence

2.8 Legal, Privacy and Ethical

ID	Requirement	Target	Verification
NFR-LEG-01	Personal data collection shall be limited to what a stated requirement needs, and the purpose shall be disclosed at the point of collection.	Data minimisation	Privacy review
NFR-LEG-02	A user shall be able to export and to erase their personal data on request.	Within 30 days	Test case
NFR-LEG-03	AI guidance on visas, health and entry requirements shall carry an explicit non-advice disclaimer.	100 percent of such output	Content review
NFR-LEG-04	Third-party content shall be used within the licence terms of its provider, and attribution shall be displayed where required.	Full compliance	Licence review
NFR-LEG-05	Trip member contact details shall not be exposed to other members beyond display name and profile photo unless the member opts in.	Opt-in only	Test case, security

3. Detailed User Stories and Use Case Diagrams

User stories express requirements from the user’s perspective and carry acceptance criteria in Given-When-Then form. Use case specifications expand the more procedurally complex interactions where the flow, not the outcome, is the difficult part.

3.1 Use Case Diagrams

Three diagrams partition the system by user journey. Actors outside the system boundary include the AI Travel Assistant, partner APIs and the payment gateway, each of which participates in use cases without being a Voyana user.

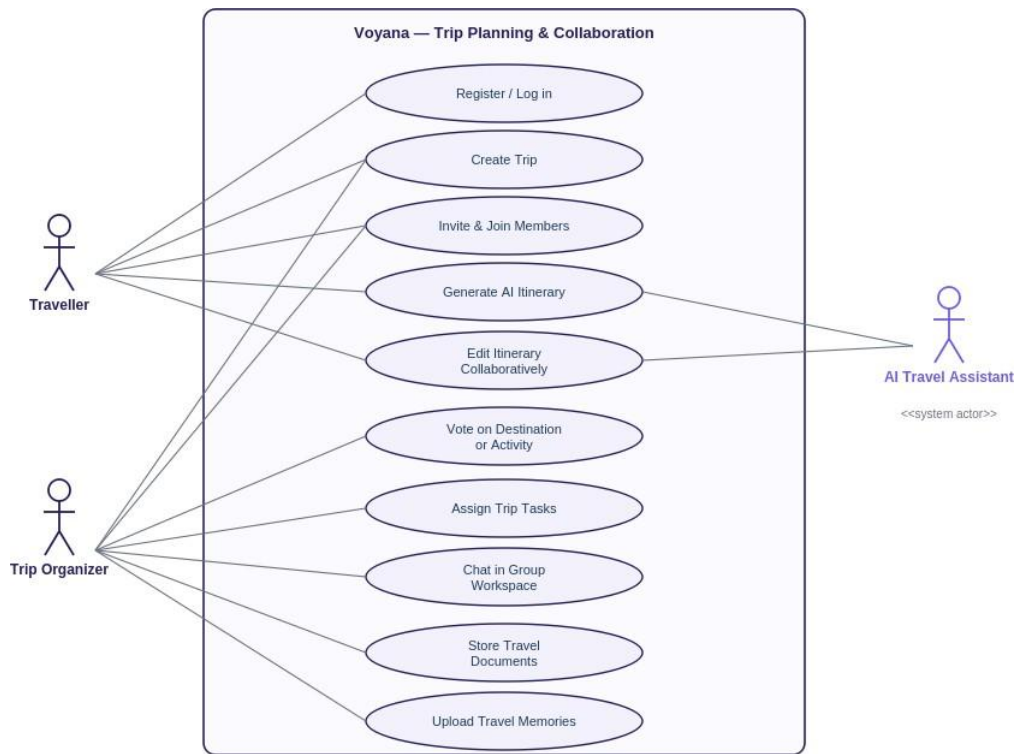


Figure 1. Use case diagram - trip planning and collaboration.

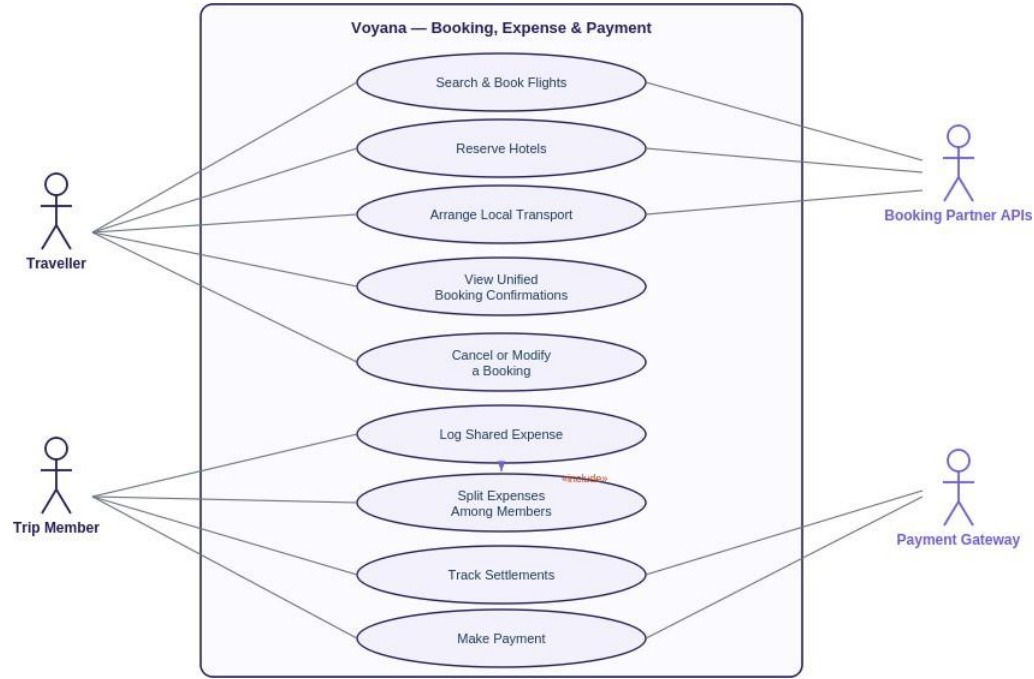


Figure 2. Use case diagram - booking, expense and payment.

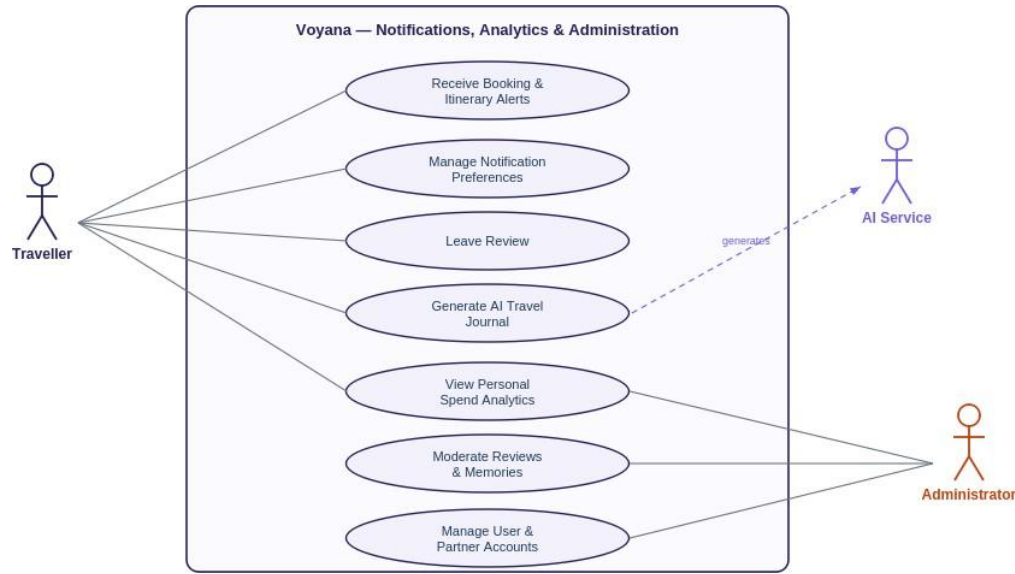


Figure 3. Use case diagram - notifications, analytics and administration.

3.2 User Stories with Acceptance Criteria

Each story carries a stable identifier, its owning epic and testable criteria. Criteria are written so that a tester who has never spoken to the author can determine pass or fail.

Field	Detail
Story	US-01 (VPM-2, 3 points)
As a	new visitor
I want	register with my email address and verify it
So that	my account and my trips are protected
Acceptance criteria	Given a valid name, an unused email and a compliant password, when I submit, then an account is created in unverified state and a six-digit code is emailed within 60 seconds. Given an email already held by an active account, when I submit, then registration is refused with a message that does not reveal whether that account exists. Given a password that fails the policy, when I type it, then strength feedback updates live and the submit control stays disabled. Given a code older than ten minutes, when I submit it, then it is rejected and I am offered a resend. Given five incorrect code attempts, when I attempt a sixth, then the code is invalidated and a new one must be requested.

Field	Detail
Story	US-02 (VPM-2, 2 points)
As a	returning traveller
I want	sign in with Google
So that	I do not maintain another password
Acceptance criteria	Given a Google account whose verified email matches an existing Voyana account, when I complete the consent flow, then I am signed in to that existing account rather than a duplicate. Given a Google account with no matching Voyana account, when I complete consent, then an account is created already verified. Given I decline consent, when I return, then I see the sign-in screen with an explanatory message and no partial account exists.

Field	Detail
Story	US-03 (VPM-4, 3 points)
As a	trip organizer
I want	create a trip with destinations, dates and a budget
So that	the group has one source of truth
Acceptance criteria	Given a title, at least one destination, a valid date range and a budget, when I submit, then the trip is created, I hold the organizer role and a day container exists for every date. Given an end date before the start date, when I submit, then creation is refused and the date field carries the error. Given a duration above 60 days, when I submit, then I receive a warning and may proceed deliberately. Given the trip is created, when I open it, then the dashboard shows zero spend against the stated budget.

Field	Detail
Story	US-04 (VPM-4, 3 points)
As a	trip organizer
I want	invite people to my trip
So that	we can plan together
Acceptance criteria	Given I am the organizer, when I invite by email, then the invitee receives an email with a link that expires after seven days. Given an invitation link, when a signed-in user opens it, then they join as a member and appear in the member list immediately. Given an expired link, when it is opened, then joining is refused with an option to request a new invitation. Given I am a member and not an organizer, when I open the members panel, then no invite control is presented and the equivalent request is refused by the service.

Field	Detail
Story	US-05 (VPM-5, 8 points)
As a	traveller
I want	have the assistant draft a day-by-day itinerary
So that	I do not spend hours researching
Acceptance criteria	Given a trip with destination, dates, budget and group size, when I request generation, then a draft covering every day with three to six activities per day returns within 30 seconds at the 95th percentile. Given a returned draft, when I review it, then each activity carries a title, start time, duration, location and estimated cost per person. Given a draft, when I have not accepted it, then the existing itinerary is unchanged. Given I accept the draft, when it is applied, then the change log records the AI Service as the actor and the applying member. Given the AI provider is unavailable, when I request generation, then I receive a clear message and manual activity entry remains available.

Field	Detail
Story	US-06 (VPM-4, 5 points)
As a	trip member
I want	edit the shared itinerary
So that	the plan reflects what the group agreed
Acceptance criteria	Given I am a member, when I add, edit, reorder or remove an activity, then all members see the change on their next load or refresh. Given another member changed the same activity after I loaded it, when I save, then my write is rejected with a conflict message showing both values. Given any change, when it is saved, then the change log records actor, timestamp and previous value. Given I am not a member, when I request the itinerary, then the service returns a forbidden response.

Field	Detail
Story	US-07 (VPM-8, 5 points)

Field	Detail
As a	trip member
I want	vote in a poll on an activity
So that	the group decides without endless messages
Acceptance criteria	Given a poll with two to six options and a closing time, when I vote, then my vote is recorded once and the tally updates for all members. Given I have already voted, when I vote again, then my earlier vote is replaced rather than added. Given the closing time passes, when any member views the poll, then it is closed, further votes are refused and the outcome is displayed. Given a closed activity poll, when an organizer promotes the winning option, then it is added to the itinerary in one action and the change log records the poll as its source.

Field	Detail
Story	US-08 (VPM-8, 5 points)
As a	trip member
I want	chat with the group inside the trip
So that	logistics stay attached to the trip
Acceptance criteria	Given I am a member and connected, when another member sends a message, then I receive it within two seconds. Given I was offline, when I reconnect, then messages sent in my absence are delivered in order without duplication. Given I am mentioned, when the message is delivered, then I receive a notification through my enabled channels. Given the trip is archived, when I open the workspace, then history is readable and the composer is disabled.

Field	Detail
Story	US-09 (VPM-6, 8 points)
As a	traveller
I want	search and book flights inside Voyana
So that	I do not lose the confirmation in my email
Acceptance criteria	Given origin, destination, dates and passenger count, when I search, then results show carrier, times, duration, stops and total price inclusive of taxes. Given a selected result, when I proceed, then the price is held for ten minutes and the remaining hold time is displayed. Given the held price has changed at confirmation, when I confirm, then the change is presented and I must accept it explicitly. Given a confirmed booking, when it completes, then it appears in the itinerary, in the unified booking view and in the expense log as one atomic action. Given the partner call fails twice, when the second retry fails, then no booking record is created and a clear failure state is shown.

Field	Detail
Story	US-10 (VPM-7, 5 points)
As a	trip member

Field	Detail
I want	log a shared expense and have it split
So that	nobody keeps a private spreadsheet
Acceptance criteria	Given an amount, currency, date, category, payer and sharing members, when I save, then the expense is recorded and each member's net position updates. Given an unequal split, when the parts do not sum to the total, then the save is refused and the shortfall is stated. Given an expense in a currency other than the trip currency, when I save, then both original and converted values are stored with the rate and date applied. Given I am neither the creator nor an organizer, when I attempt to edit or delete it, then the action is refused.

Field	Detail
Story	US-11 (VPM-7, 5 points)
As a	trip member
I want	see who owes whom and settle up
So that	the trip ends without argument
Acceptance criteria	Given all logged expenses, when I open settlements, then the obligations are reduced to the minimum number of transfers. Given the settlement set, when every transfer is summed, then all member positions reconcile to zero. Given I mark a transfer as paid, when the receiving member confirms, then both positions update and the transfer is recorded with a timestamp. Given the receiving member has not confirmed, when either party views it, then the transfer shows as pending confirmation.

Field	Detail
Story	US-12 (VPM-5, 5 points)
As a	traveller
I want	ask the assistant questions about my trip
So that	I get answers about my plan, not generic search results
Acceptance criteria	Given a trip with an itinerary and a budget, when I ask a trip-specific question, then the answer references the actual itinerary or budget values. Given the assistant proposes a change, when it is presented, then no change is applied until I accept it. Given the AI provider is rate limited, when I send a message, then I am told so and invited to retry. Given any assistant response, when it is displayed, then it is visibly marked as AI-generated.

Field	Detail
Story	US-13 (VPM-9, 3 points)
As a	traveller
I want	store my passport and visa securely
So that	I can retrieve them when I travel

Field	Detail
Acceptance criteria	Given a PDF, JPEG or PNG under 10 MB, when I upload it, then it is stored encrypted and appears in the trip vault. Given an uploaded document, when I open it, then it is served through a signed URL valid for no more than 15 minutes. Given I have not shared a document, when another member opens the vault, then that document is not listed. Given an expiry date, when the document is within 180, 90 or 30 days of it, then I receive a reminder.

Field	Detail
Story	US-14 (VPM-9, 3 points)
As a	traveller
I want	attach photos to the day they happened
So that	the album keeps its context
Acceptance criteria	Given photographs with capture timestamps inside the trip range, when I upload them, then each is assigned to the matching itinerary day automatically. Given a photograph without a usable timestamp, when I upload it, then I am prompted to choose a day before the upload completes. Given the album, when I open it, then items are grouped by day with the associated activity shown. Given I delete my own item, when the deletion completes, then it is removed from storage within 24 hours.

Field	Detail
Story	US-15 (VPM-9, 3 points)
As a	traveller
I want	control which notifications reach me
So that	the application is useful rather than noisy
Acceptance criteria	Given the preference screen, when I disable a category for push, then no further push of that category is delivered while the in-application entry is still created. Given push permission is absent, when an event fires, then it is delivered to the notification centre instead. Given the same event fires twice within five minutes, when the second fires, then no duplicate is delivered. Given the event behind a scheduled reminder is deleted, when the reminder time arrives, then no reminder is sent.

Field	Detail
Story	US-16 (VPM-9, 3 points)
As a	administrator
I want	moderate reported content
So that	the platform stays safe
Acceptance criteria	Given a reported review or memory, when I open the queue, then it is listed with the reporting reason and the reporting count.

Field	Detail
	Given a queued item, when I approve or remove it, then a reason is required and the decision is recorded in the audit log with my identity and the timestamp. Given I remove an item, when the author next opens it, then they see that it was removed and why. Given I am not an administrator, when I request the queue endpoint, then the service returns a forbidden response.

A further ten stories covering trip templates, shared calendar, packing lists, destination recommendation, journal generation, session management, data export, refunds, booking amendment and personal spend analytics are maintained in Jira under the same identifier scheme and are not reproduced here.

3.3 Expanded Use Case Specifications

UC-01 Generate and Accept an AI Itinerary

Field	Detail
Use case	UC-01 - Generate and Accept an AI Itinerary
Primary actor	Trip member
Supporting actors	AI Travel Assistant, Trip Service, Weather provider
Preconditions	1. The member is authenticated and belongs to the trip. 2. The trip carries a destination, a date range and a budget.
Postconditions	1. A draft itinerary exists or the itinerary is unchanged. 2. Every applied change is recorded in the change log.
Main success scenario	1. The member opens the itinerary and requests generation. 2. The system assembles trip context: destinations, dates, budget, group size and member preferences. 3. The system calls the AI Service and displays a progress indicator. 4. The AI Service returns a draft of three to six activities per day with times, durations and per-person costs. 5. The system displays the draft marked as AI-generated alongside the existing itinerary. 6. The member reviews, edits individual activities and accepts the draft. 7. The system writes the activities, recomputes the estimated trip cost and records the change. 8. The system notifies the other members that the itinerary changed.
Alternative flows	4a. The estimated cost exceeds the trip budget: the system flags the overage and offers a lower-cost regeneration. 6a. The member rejects the draft: nothing is written and the existing itinerary stands. 6b. The member accepts only selected days: only those days are written.
Exception flows	3a. The AI provider times out after 30 seconds: the system reports the failure, offers a retry and keeps manual entry available. 3b. The provider returns a rate-limit response: the system states when a retry may be attempted. 7a. A concurrent edit has changed the itinerary: the system reports the conflict and asks the member to review before writing.

UC-02 Book a Flight for the Trip

Field	Detail
Use case	UC-02 - Book a Flight for the Trip
Primary actor	Trip organizer
Supporting actors	Booking partner API, Payment gateway, Expense Service
Preconditions	1. The organizer is authenticated and holds the organizer role on the trip. 2. The trip has dates and at least one member.
Postconditions	1. Either a confirmed booking exists with a linked expense entry, or no partial record exists. 2. Every attempt is recorded with its outcome.
Main success scenario	1. The organizer enters origin, destination, dates and passenger count and searches. 2. The Booking Service queries the partner adapter and returns results with price, duration, stops and cancellation terms. 3. The organizer selects a result; the system holds the price for ten minutes and displays the remaining time.

Field	Detail
	4. The system displays the total and the per-person split. 5. The organizer proceeds to payment; card capture occurs in the gateway-hosted component. 6. The gateway reports success and sends a webhook. 7. The system confirms the booking with the partner, writes it to the itinerary, the unified booking view and the expense log in one transaction. 8. The system notifies all members.
Alternative flows	3a. The hold expires before payment: the system re-prices and asks the organizer to confirm the new price. 5a. The organizer abandons payment: the hold lapses and no record is created. 7a. A duplicate webhook arrives: the idempotency key causes it to be ignored.
Exception flows	2a. The partner returns an error or times out: the system retries twice with backoff, then reports failure without creating a record. 6a. The gateway declines: the booking is not confirmed, the cart is preserved and the organizer may retry. 7a. The partner fails after successful payment: the system marks the booking as requiring intervention, raises an administrator alert and initiates a refund under FR-PAY-06.

UC-03 Split an Expense and Settle Up

Field	Detail
Use case	UC-03 - Split an Expense and Settle Up
Primary actor	Trip member
Supporting actors	Expense Service, Payment gateway, Currency provider
Preconditions	1. The member is authenticated and belongs to the trip. 2. The trip has at least two members.
Postconditions	1. Net positions reflect every logged expense and every confirmed settlement. 2. All positions reconcile to zero.
Main success scenario	1. The member logs an expense with amount, currency, date, category, payer and sharing members. 2. The system converts to the trip currency using the rate for the expense date and stores both values with the rate. 3. The system computes the split and updates every affected net position. 4. Any member opens settlements; the system reduces obligations to the minimum transfer set. 5. A debtor marks a transfer as paid, optionally initiating payment through the gateway in test mode. 6. The creditor confirms receipt. 7. The system records the settlement and updates both positions.
Alternative flows	1a. An unequal split is chosen: the member enters amounts or percentages and the system validates that they reconcile. 5a. Settlement occurs outside the platform: the debtor records it manually and the creditor still confirms.
Exception flows	2a. The currency provider is unavailable: the system uses the most recent cached rate, marks the expense as provisionally converted and re-converts when the provider returns. 3a. The split does not reconcile: the save is refused and the shortfall is stated. 6a. The creditor disputes the transfer: it remains pending and both members are notified.

UC-04 Run a Group Poll and Apply the Outcome

Field	Detail
Use case	UC-04 - Run a Group Poll and Apply the Outcome
Primary actor	Trip member
Supporting actors	Trip organizer, Notification Service, Trip Service
Preconditions	1. The member is authenticated and belongs to the trip. 2. The trip has at least two members.
Postconditions	1. The poll is closed with a recorded outcome and closing actor. 2. If promoted, the winning option exists as an itinerary activity.
Main success scenario	1. The member creates a poll with a question, two to six options and a closing time. 2. The system notifies all members through their enabled channels. 3. Members vote; the system records one vote each and updates tallies live. 4. The closing time passes or an organizer closes the poll early. 5. The system records the outcome, the closing actor and the timestamp, and notifies all members. 6. An organizer promotes the winning option to the itinerary. 7. The system creates the activity and records the poll as its source in the change log.
Alternative flows	3a. A member changes their vote before closing: the earlier vote is replaced. 4a. The result is tied: the system reports the tie and requires the organizer to select among the tied options, recording the manual decision. 6a. The outcome is not promoted: the poll remains closed as a record only.
Exception flows	3a. A non-member attempts to vote: the service returns a forbidden response. 6a. The itinerary has changed such that the day no longer exists: the system asks the organizer to select a different day.

UC-05 Register and Verify an Account

Field	Detail
Use case	UC-05 - Register and Verify an Account
Primary actor	Visitor
Supporting actors	Auth Service, Email provider
Preconditions	1. The visitor is not authenticated. 2. The visitor holds an email address they control.
Postconditions	1. Either a verified account exists, or no usable account exists. 2. No password is stored in recoverable form.
Main success scenario	1. The visitor submits name, email and password. 2. The system validates the password policy and the uniqueness of the email. 3. The system creates an unverified account, hashes the password with bcrypt and dispatches a six-digit code. 4. The visitor submits the code within ten minutes. 5. The system marks the account verified, issues an access and refresh token pair and opens the dashboard.
Alternative flows	1a. The visitor chooses Google or Apple sign-in: the account is created already verified and steps 3 and 4 are skipped. 4a. The code expires: the visitor requests a new one; the previous code is invalidated.

Field	Detail
Exception flows	2a. The email is already registered: the system refuses without disclosing whether the account exists and offers password reset. 3a. Email dispatch fails: the system retries three times, then surfaces the failure and offers an alternative address. 4a. Five incorrect attempts occur: the code is invalidated and a new one must be requested.

UC-06 Moderate Reported Content

Field	Detail
Use case	UC-06 - Moderate Reported Content
Primary actor	Administrator
Supporting actors	Analytics Service, Notification Service
Preconditions	1. The administrator is authenticated with the administrator role. 2. At least one item is reported or automatically flagged.
Postconditions	1. Every queued item is approved or removed with a recorded reason. 2. Each decision is present in the immutable audit log.
Main success scenario	1. The administrator opens the moderation queue. 2. The system lists items with the reporting reason, the reporting count and the flag source. 3. The administrator opens an item and reviews it in context. 4. The administrator approves or removes it and records a reason. 5. The system applies the decision, writes an audit entry and notifies the author when the item is removed.
Alternative flows	3a. The item was already actioned by another administrator: the system reports this and removes it from the queue. 4a. The administrator escalates: the item is flagged for review by the project lead and remains queued.
Exception flows	1a. A non-administrator requests the queue: the service returns a forbidden response. 5a. The audit write fails: the decision is rolled back and an error is raised, so that no moderation action exists without an audit record.

4. Interface Requirements

Interface requirements define every boundary the platform presents or consumes: to the user, to other software, to hardware, over the network and to external partners. Identifiers use the form IR-<TYPE>-<NN>.

4.1 System Context

The responsive web client is the sole user-facing interface. It communicates only with the API gateway, which routes to twelve independently deployable services. No service is addressed directly by the browser, and no service reaches into another service’s datastore.

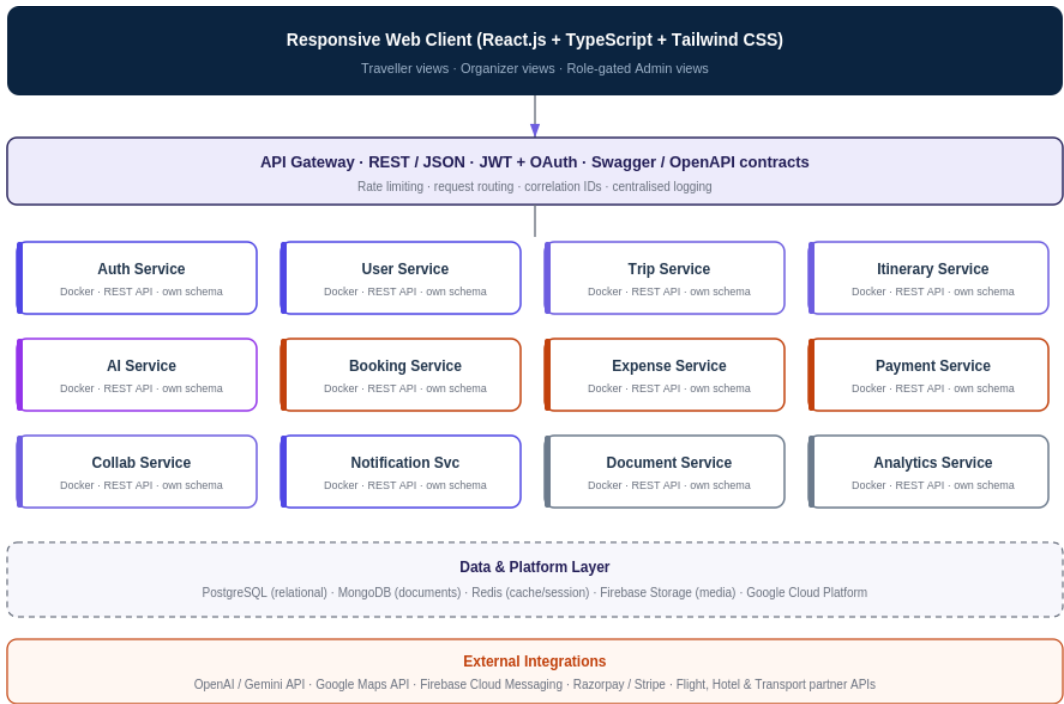


Figure 5. Voyana system context and service decomposition.

4.2 User Interface Requirements

ID	Requirement	Applies to
IR-UI-01	The interface shall use a single design system covering colour, typography, spacing, iconography and component states, published in Figma and implemented as reusable React components.	All screens
IR-UI-02	The primary navigation shall expose Dashboard, My Trips, Itinerary, Bookings, Expenses, Workspace, Documents and Memories, and shall indicate the active section at all times.	Authenticated shell
IR-UI-03	The layout shall adapt across four breakpoints, collapsing the sidebar to a bottom navigation bar below 768 pixels.	All screens
IR-UI-04	Every form field shall carry a persistent label, and validation shall appear inline against the field that failed, not as a summary alone.	All forms
IR-UI-05	Monetary values shall display with an explicit currency symbol and thousands separators appropriate to the trip currency.	Expenses, Bookings, Dashboard
IR-UI-06	Loading, empty, error and success states shall be designed explicitly for every data-bearing view; a blank region is not an acceptable empty state.	All data views
IR-UI-07	AI-generated content shall carry a visible marker and a control to regenerate or to dismiss it.	Itinerary, Assistant, Journal

ID	Requirement	Applies to
IR-UI-08	Colour shall never be the sole carrier of meaning; status shall also be conveyed by text or shape.	All screens
IR-UI-09	All interactive elements shall be reachable by keyboard in a logical order with a visible focus indicator.	All screens
IR-UI-10	Destructive actions shall use a confirmation dialogue that names the object and states the consequence.	All destructive actions

4.3 Software Interfaces

ID	Requirement	Applies to
IR-SW-01	All inter-service communication shall use REST over HTTPS with JSON request and response bodies.	All 12 services
IR-SW-02	Every service shall publish an OpenAPI 3.0 document at /docs, and the contract shall be agreed before implementation begins.	All 12 services
IR-SW-03	Endpoints shall be versioned in the path as /api/v1/<resource>, and a breaking change shall require /api/v2.	All 12 services
IR-SW-04	Error responses shall use a single envelope containing a machine-readable code, a human-readable message, an optional field-level detail array and the correlation identifier.	All 12 services
IR-SW-05	Collection endpoints shall support cursor pagination, sorting and filtering, and shall never return an unbounded collection.	All collection endpoints
IR-SW-06	Authentication shall be carried as a bearer token in the Authorization header; no service shall accept credentials in a query string.	All 12 services
IR-SW-07	Every request shall carry an X-Correlation-Id header, generated at the gateway when absent and propagated unchanged across all downstream calls.	All 12 services
IR-SW-08	Write endpoints that can be retried shall accept an Idempotency-Key header and shall return the original result for a repeated key.	Booking, Payment, Expense
IR-SW-09	Each service shall own its schema; cross-service data access shall occur only through the owning service's interface.	All 12 services
IR-SW-10	Redis shall be used for session and read-through caching only; it shall never be the system of record.	Auth, Trip, Booking

4.3.1 Representative Endpoint Contract

Method and path	Purpose	Success	Principal errors
POST /api/v1/auth/register	Create an unverified account and dispatch a verification code.	201 Created	409 email in use, 422 validation
POST /api/v1/auth/login	Exchange credentials for an access and refresh token pair.	200 OK	401 invalid, 423 locked
POST /api/v1/trips	Create a trip and its day containers.	201 Created	422 date range invalid
GET /api/v1/trips/{id}/itinerary	Retrieve the itinerary with an entity tag for concurrency control.	200 OK	403 not a member, 404 unknown trip
PATCH /api/v1/trips/{id}/activities/{aid}	Update an activity under optimistic locking.	200 OK	409 stale version, 403 not a member

Method and path	Purpose	Success	Principal errors
POST /api/v1/ai/itinerary	Generate a draft itinerary from trip context.	202 Accepted	429 rate limited, 503 provider down
POST /api/v1/bookings/flights/search	Search partner flight inventory.	200 OK	502 partner error, 504 partner timeout
POST /api/v1/bookings/{id}/confirm	Confirm a held booking and write the linked expense.	201 Created	402 payment failed, 409 price changed
POST /api/v1/expenses	Log an expense and compute the split.	201 Created	422 split does not reconcile
GET /api/v1/trips/{id}/settlements	Return the simplified settlement set.	200 OK	403 not a member
POST /api/v1/payments/webhook	Receive a gateway notification; idempotent.	200 OK	400 signature invalid

4.4 Hardware and Platform Interfaces

ID	Requirement	Applies to
IR-HW-01	The client shall run in a standard browser on desktop, tablet and smartphone hardware; no dedicated device is required.	Client
IR-HW-02	The client shall request camera or gallery access only when the user initiates an upload of a document, receipt or memory.	Client
IR-HW-03	The client shall request geolocation only when the user invokes a location-dependent feature, and shall function fully without it.	Client
IR-HW-04	All services shall run as containers on Google Cloud Platform with no dependency on a specific host instance.	Platform
IR-HW-05	Media shall be stored in object storage and shall be served through signed URLs rather than from application servers.	Platform

4.5 External Interfaces

Every external dependency is reached through an adapter that isolates the provider's contract from Voyana's domain model, so that a provider can be replaced without altering the calling service.

ID	External system	Direction and protocol	Voyana requirement
IR-EX-01	OpenAI or Gemini API	Outbound HTTPS, JSON	Requests carry no personally identifying data beyond what a stated requirement needs. Timeout 30 s, two retries, circuit breaker after five consecutive failures.
IR-EX-02	Google Maps API	Outbound HTTPS, JSON	Used for geocoding, place detail and route display. Results cached for 24 hours to remain inside quota.
IR-EX-03	Flight, hotel and transport partner APIs	Outbound HTTPS, JSON	Accessed only through the booking adapter interface. Sandbox this release; contract identical for a commercial provider.
IR-EX-04	Razorpay or Stripe	Outbound HTTPS plus inbound webhook	Card capture occurs in the hosted component only. Webhooks are signature-verified and processed idempotently.
IR-EX-05	Firebase Cloud Messaging	Outbound HTTPS	Web push delivery. Failure falls back to the in-application notification centre.

ID	External system	Direction and protocol	Voyana requirement
IR-EX-06	Firebase Storage	Outbound HTTPS	Media and document storage. Access exclusively through signed URLs valid for at most 15 minutes.
IR-EX-07	Transactional email provider	Outbound SMTP or HTTPS	Verification codes, password reset links and settlement requests. Delivery failures are retried three times and then surfaced in the notification centre.
IR-EX-08	Currency rate provider	Outbound HTTPS, JSON	Daily rate retrieval. The rate applied to an expense is stored with the expense so that historical values never change.

4.6 Communication Interfaces

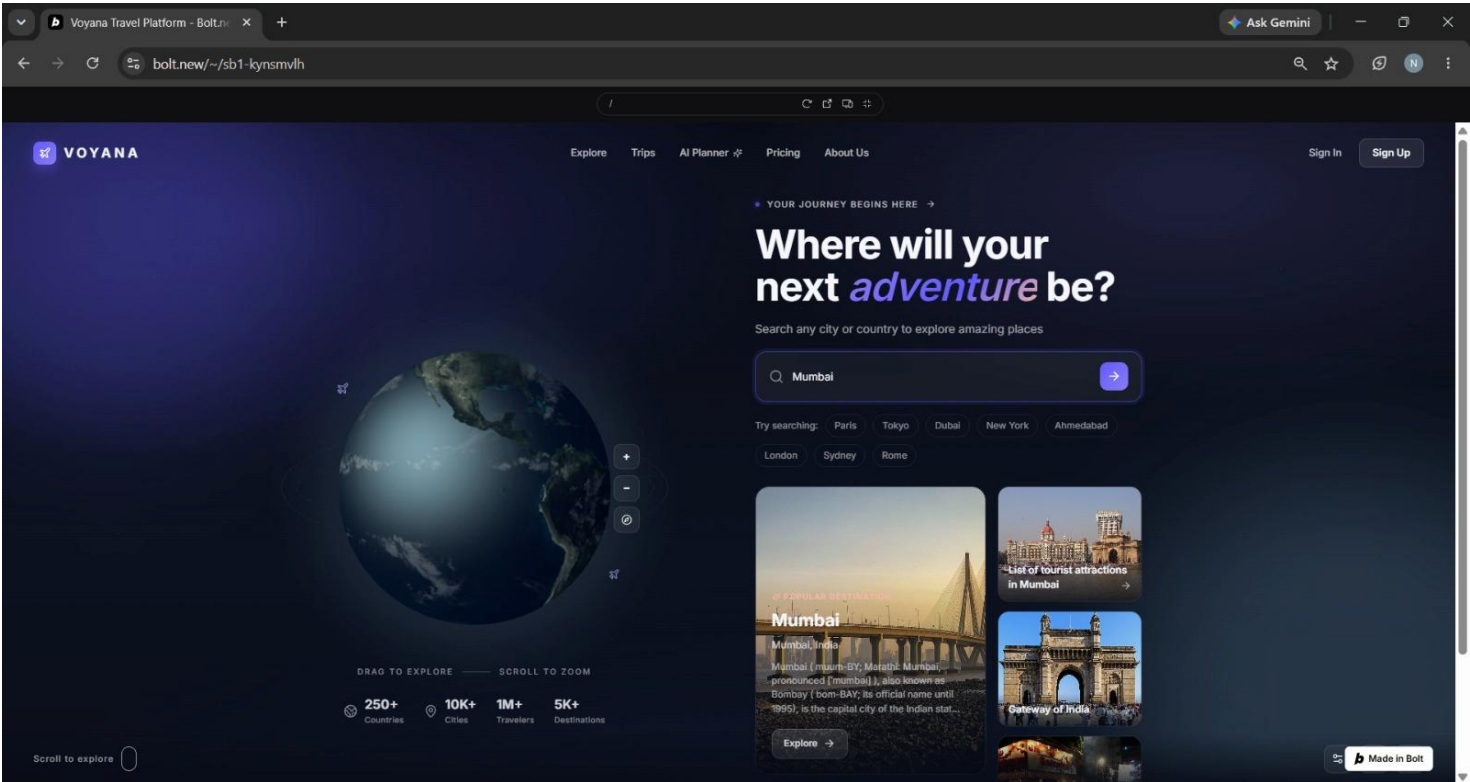
- All external traffic uses HTTPS with TLS 1.2 or higher; plaintext HTTP is redirected and never served.
- Group chat and workspace presence use a WebSocket channel authenticated by the same JWT as the REST interface, with automatic reconnection and message replay from the last acknowledged identifier.
- Web push uses the Firebase Cloud Messaging transport; the payload carries an event identifier and a deep link only, never trip content.
- Inbound webhooks are accepted only from allow-listed origins and only with a valid provider signature.
- Cross-origin resource sharing is restricted to the published client origins; wildcard origins are not permitted in any environment

5. Wireframe- UI/UX Mockups

5.1 Public Landing Page

The unauthenticated entry point. It states the value proposition, shows a live itinerary card as proof rather than description, and offers a single primary action. The four-item feature strip maps directly to the four MVP pillars: AI planning, unified booking, fair splitting and contextual memories.

Figure 2. Public Landing Page - first-draft high-fidelity mockup.



Requirements satisfied	Open design decisions
IR-UI-01, IR-UI-03, IR-UI-08, NFR-USA-02	Whether the demonstration video is hosted or embedded; whether pricing is shown at all for an academic release.

5.2 Registration

Registration presents federated sign-in first and the email path second, because the majority of the UAT panel is expected to hold a Google account. Password strength is shown live, the terms checkbox is explicit and unticked by default, and the verification-code expiry is stated before the user submits rather than after.

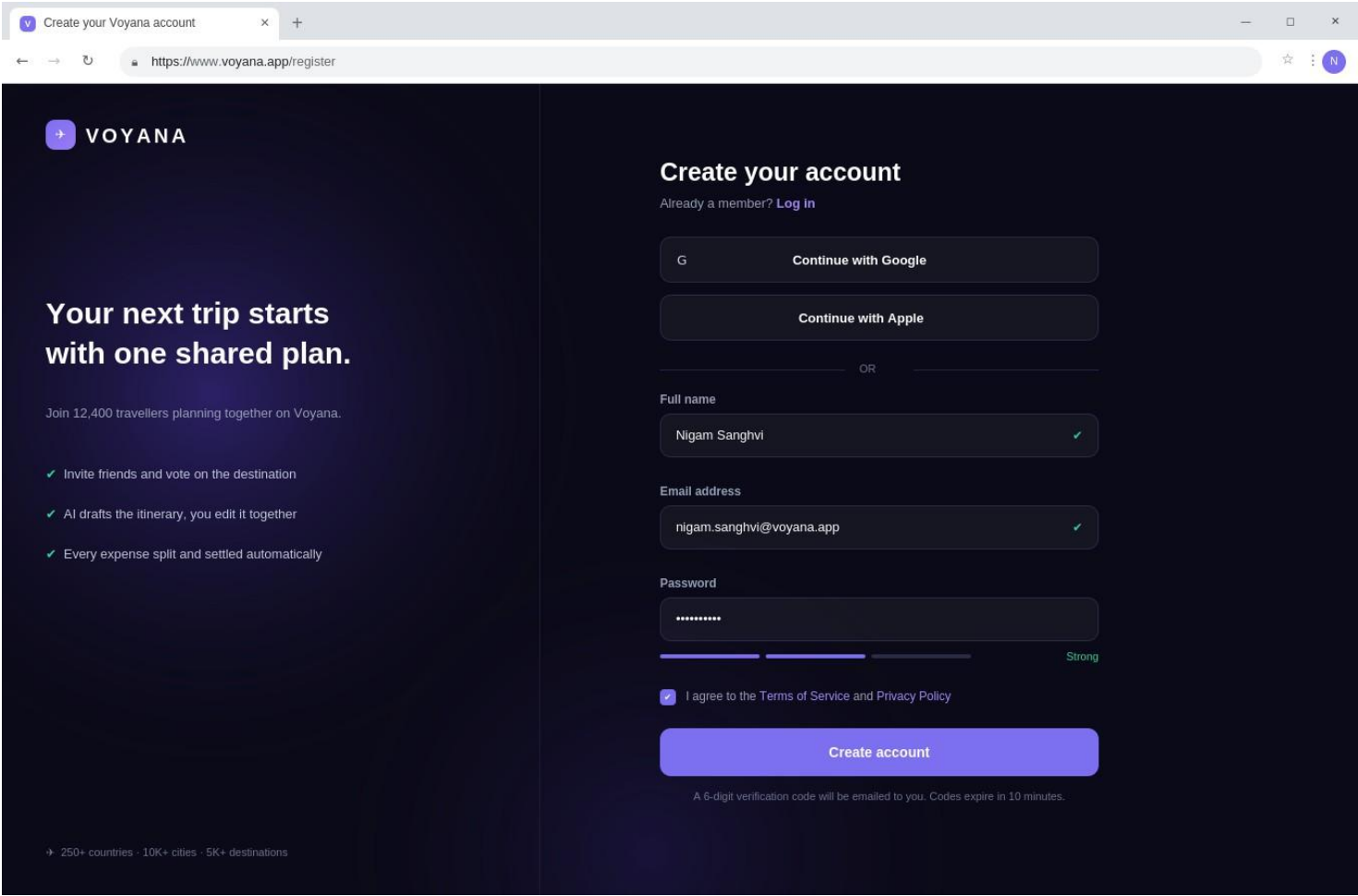


Figure 3. Registration - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-AUT-01, FR-AUT-02, FR-AUT-03, FR-AUT-05, IR-UI-04, NFR-USA-05	Whether Apple sign-in is retained given the additional configuration cost; whether the terms checkbox blocks submission or is validated on submit.

5.3 Trip Dashboard

The authenticated home. Trip cards carry status, dates, membership and planning progress. The metric strip answers the four questions a traveller opens the application to ask. The AI suggestion panel is deliberately advisory: every card proposes a change and requires an explicit action, satisfying the requirement that AI never mutates a plan silently.

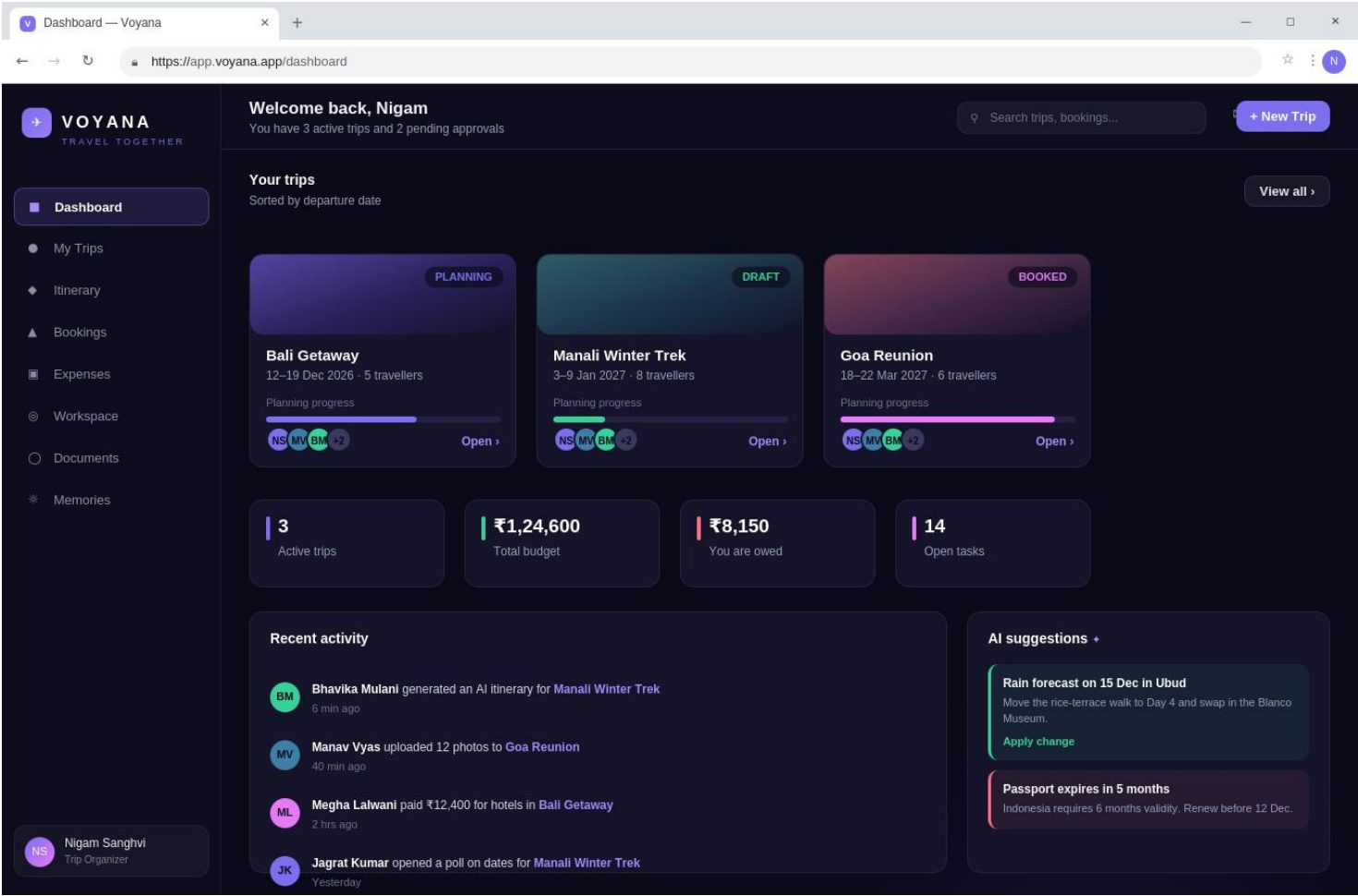


Figure 4. Trip Dashboard - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-TRP-06, FR-AI-07, FR-DOC-04, FR-EXP-04, IR-UI-02, IR-UI-07	Whether trip cards should surface unsettled balance directly; how the dashboard degrades when a user has no trips at all.

5.4 Itinerary Builder with AI Assistant

The core working screen. The day selector, the activity list and the assistant occupy one view so that a conversational instruction and its result are visible together. Each activity row carries time, duration, location, category and cost, and exposes edit and drag handles. The assistant states the cost consequence of every change it proposes.

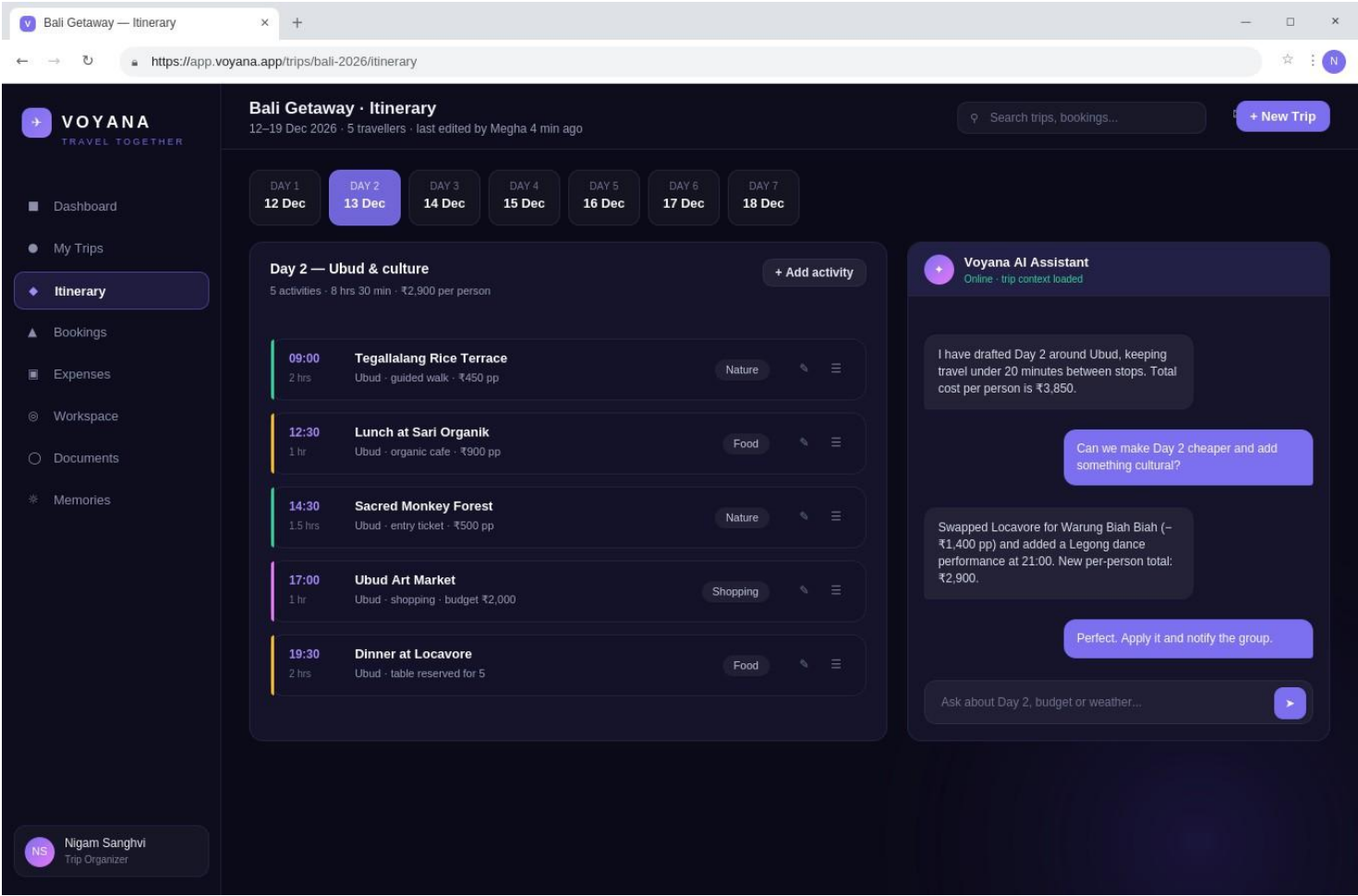


Figure 5. Itinerary Builder with AI Assistant - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-TRP-03, FR-TRP-04, FR-AI-01, FR-AI-02, FR-AI-04, FR-AI-12, IR-UI-07	Whether drag reordering is available on touch at 360 pixels or replaced by explicit move controls; whether the assistant collapses to a launcher below 1024 pixels.

5.5 Group Workspace

Chat, polls and tasks in one view, because separating them is what fragments group decisions in the tools Voyana replaces. The poll panel shows live tallies, a closing deadline and a single action that promotes the winning option into the itinerary. Tasks show assignee and completion state without a separate screen.

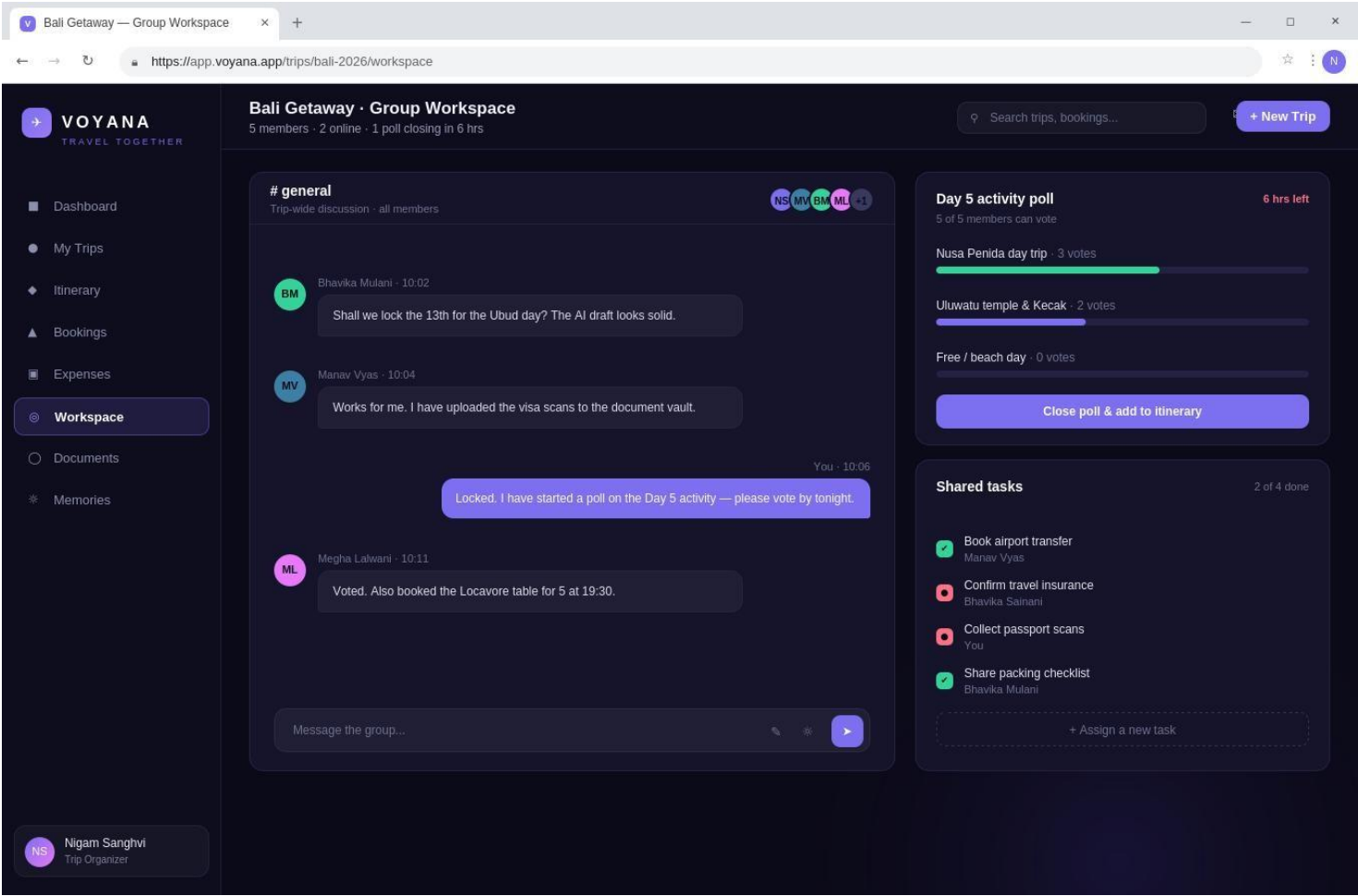


Figure 6. Group Workspace - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-COL-01, FR-COL-02, FR-COL-04, FR-COL-05, FR-COL-06, FR-COL-08, NFR-PRF-04	Whether the workspace supports channels beyond a single general channel in the MVP; how long chat history is retained after a trip is archived.

5.6 Expenses and Settlements

Budget consumption sits above the log so that the consequence is visible before the detail. Every expense states its payer and its split. The settlement panel presents the simplified transfer set rather than the raw obligation graph, which is the difference between a usable feature and a spreadsheet.

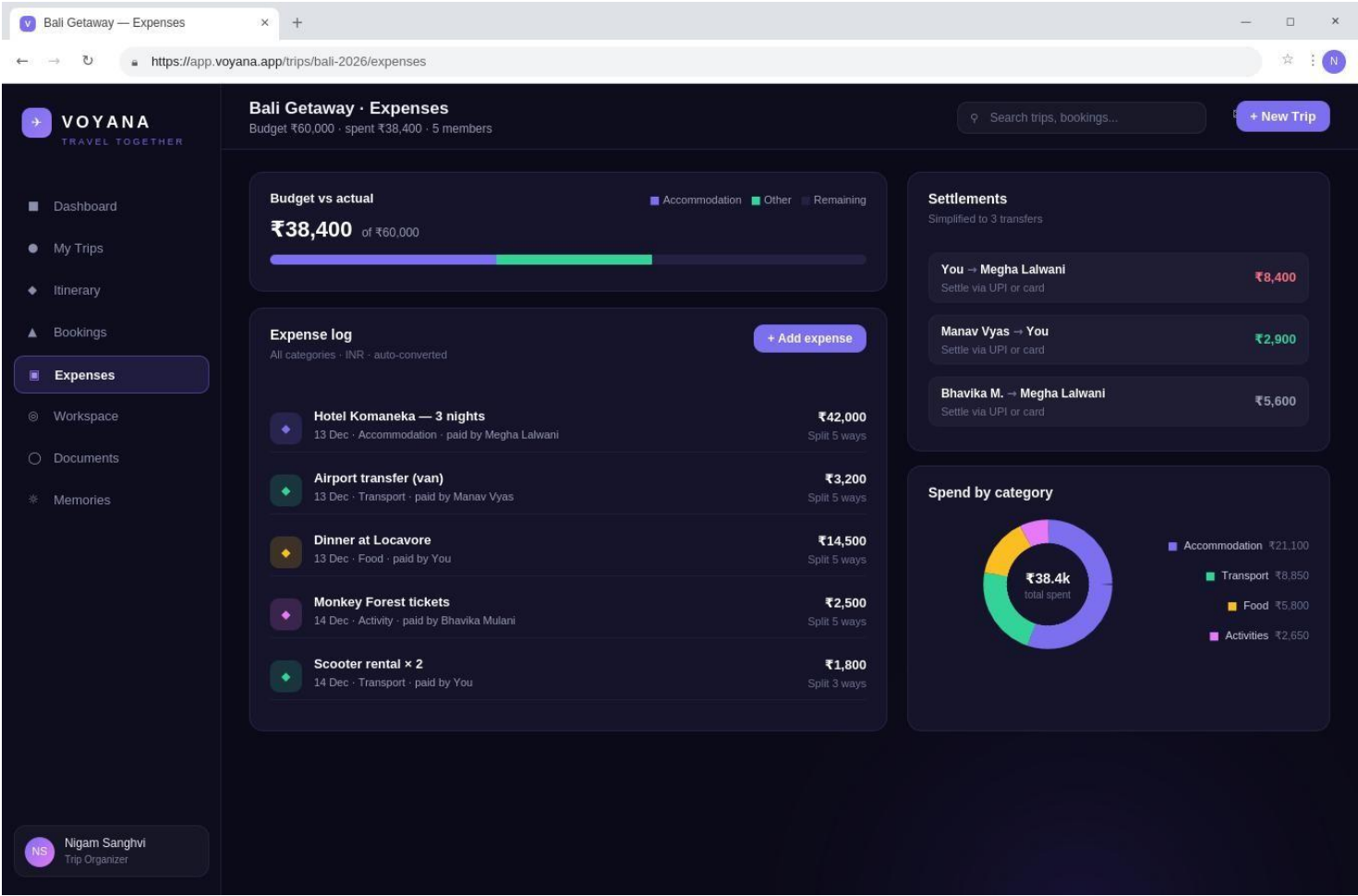


Figure 7. Expenses and Settlements - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-EXP-01, FR-EXP-02, FR-EXP-03, FR-EXP-04, FR-EXP-05, FR-EXP-06, IR-UI-05	Whether unequal splits are edited inline or in a dedicated dialogue; whether settlement initiation is restricted to the debtor.

5.7 Unified Booking

Search criteria, the result set and the running summary occupy one view. The summary states the per-person split before payment, which is the point at which a group booking usually stalls. Cancellation terms appear against each result rather than behind a link, because that is the information travellers report losing.

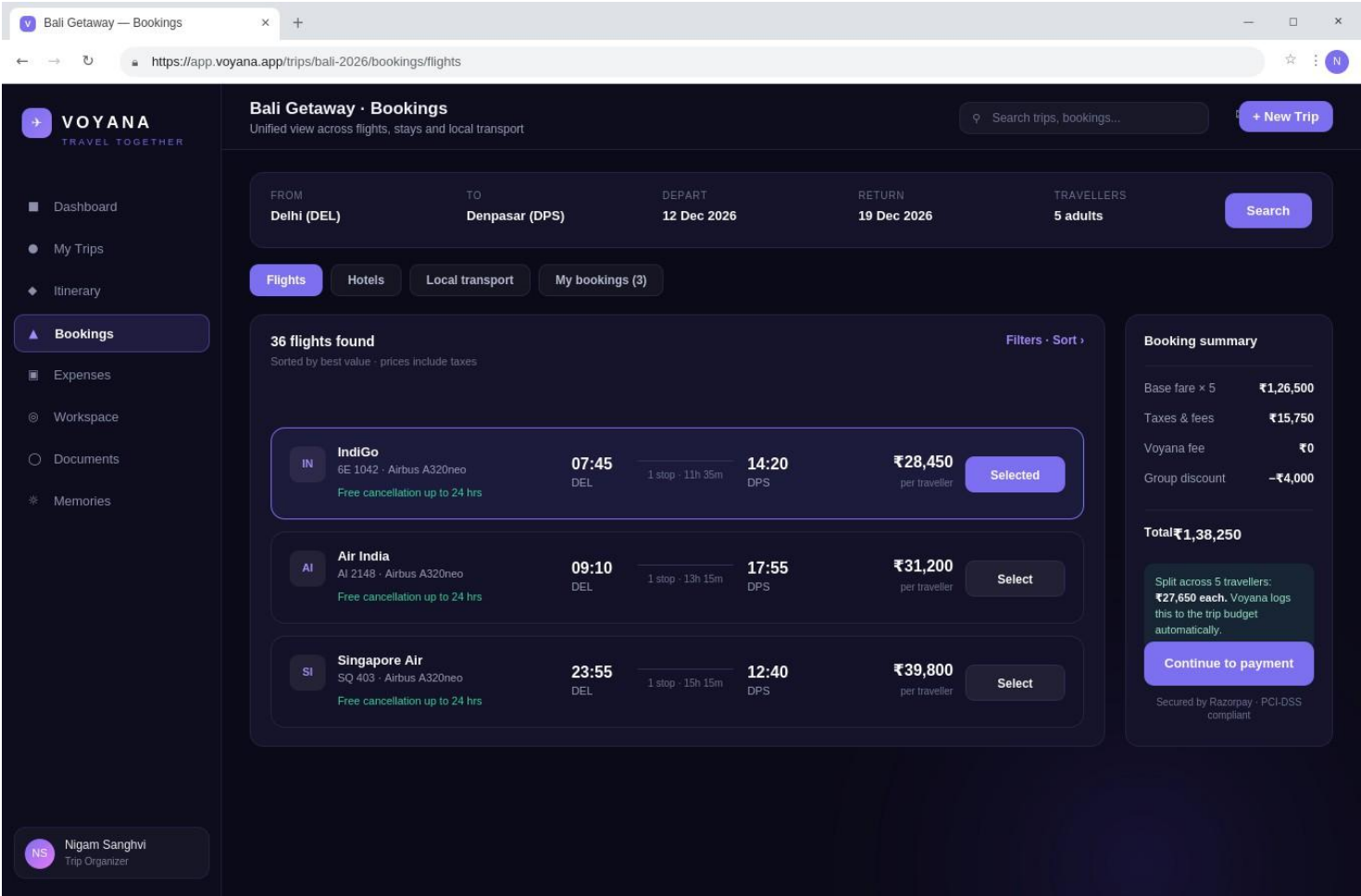


Figure 8. Unified Booking - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-BKG-01, FR-BKG-04, FR-BKG-05, FR-BKG-08, FR-BKG-09, FR-PAY-01, IR-UI-05	Whether hotels and transport reuse this layout or require distinct result cards; how a held price expiring mid-session is communicated.

5.8 Administration and Analytics Console

The administrator view, delivered at demonstration depth. It combines platform metrics, a service health panel that reflects the twelve-service architecture, and a moderation queue where every action records a reason. Figures shown are seeded, since no production traffic exists during the delivery window.

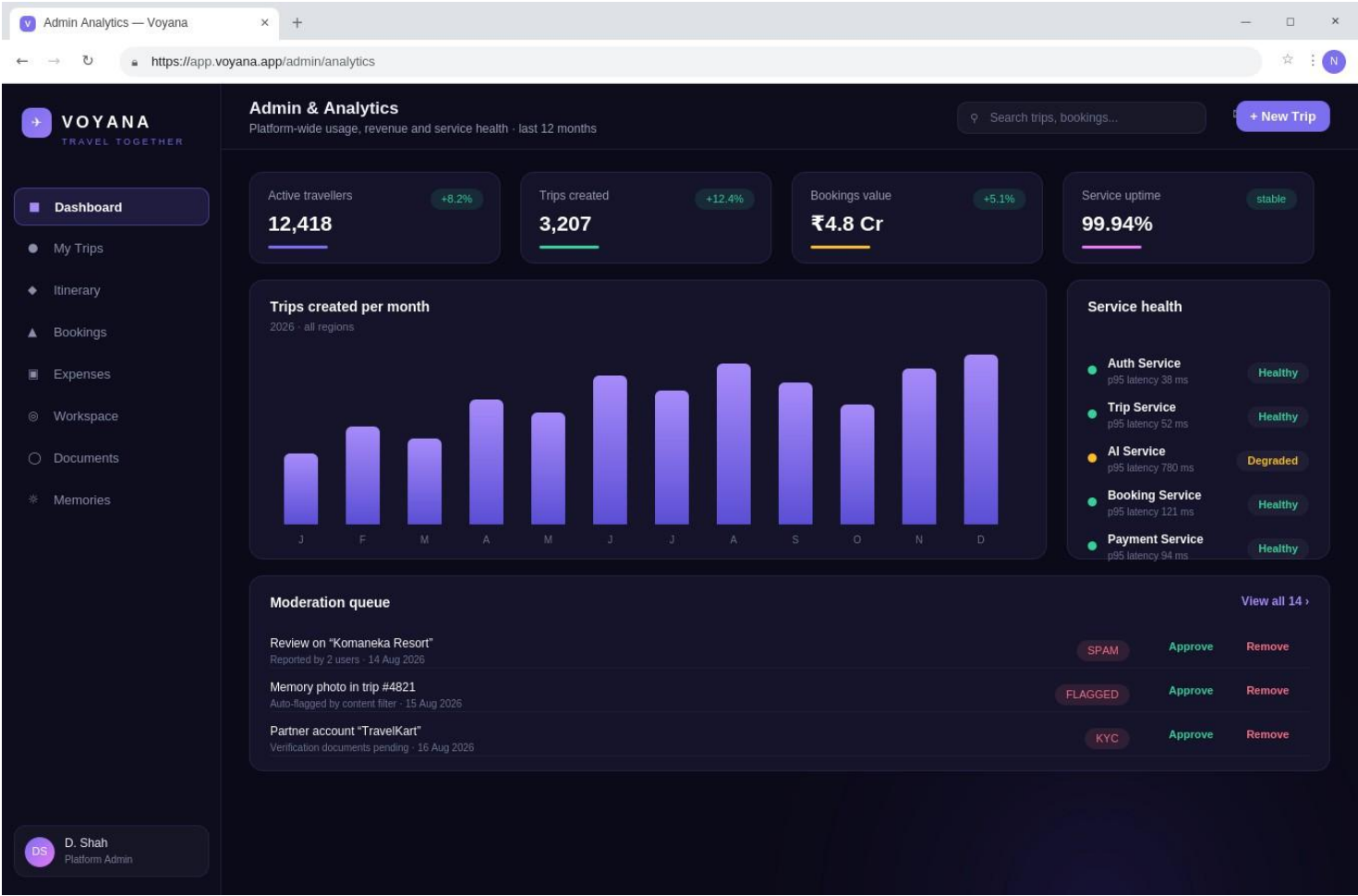


Figure 9. Administration and Analytics Console - first-draft high-fidelity mockup.

Requirements satisfied	Open design decisions
FR-ADM-01, FR-ADM-02, FR-ADM-03, FR-ADM-04, FR-ADM-07	Whether service health is read live from health endpoints or polled by the Analytics Service; what retention applies to the moderation audit log.

5.9 Responsive Behaviour

Each screen above is specified at four breakpoints. The table records the adaptation rule rather than a separate mockup for each width.

Screen	360 px (mobile)	768 px (tablet)	1024 px and above (desktop)
Dashboard	Trip cards stack; metric strip becomes a two-column grid; activity feed collapses to five items.	Trip cards in two columns; metrics in four columns.	As mocked up.
Itinerary	Assistant collapses to a floating launcher; day selector becomes a horizontal scroller.	Assistant becomes a bottom sheet; activity list full width.	As mocked up, two columns.

Screen	360 px (mobile)	768 px (tablet)	1024 px and above (desktop)
Workspace	Chat, polls and tasks become three tabs.	Chat full width; polls and tasks in an accordion below.	As mocked up, two columns.
Expenses	Budget bar pinned to top; log as compact rows; settlements in a tab.	Log full width; settlements and chart below.	As mocked up, two columns.
Booking	Criteria collapse to a summary chip; summary becomes a sticky footer bar.	Results full width; summary as a sticky footer.	As mocked up, two columns.
Admin	Not supported below 768 px; a message directs the administrator to a larger display.	Metrics stack; moderation queue as cards.	As mocked up.

6. Acceptance Criteria

Acceptance operates at three levels: each requirement carries its own verification method in Sections 1 and 2; each module carries an exit condition below; and the release as a whole carries a Definition of Done and a user acceptance plan.

6.1 Module Acceptance Criteria

Module	Accepted when	Evidence
Authentication and User Management	A user can register, verify, sign in by both paths, reset a password, revoke a session and deactivate; every trip-scoped endpoint refuses a non-member; account lockout triggers at five failures.	Test report, security suite
Trip Planning	A trip can be created, populated, edited, templated and shared; date changes re-map days correctly; conflicting writes are rejected; the change log is complete for every edit.	Test report, change log export
AI Services	Generation returns within 30 seconds at the 95th percentile across 20 runs; no AI output mutates a plan without acceptance; every AI screen retains a manual path with the provider disabled.	Timing log, fault injection report
Collaboration	Chat delivers within two seconds; polls accept one vote per member and close correctly including the tie path; tasks assign and complete; the calendar reflects all three sources.	Test report, instrumented timings
Booking Services	Search, hold, confirm, view and cancel complete against the sandbox; a confirmed booking always produces a linked expense; two induced partner failures produce no partial records.	Integration report, fault injection report
Expense Management	Equal and unequal splits validate correctly; settlements reconcile to zero on all seeded datasets; multi-currency expenses store rate and date; audit log covers every edit and deletion.	Automated test output
Payment	Test-mode payment confirms a booking; duplicate webhooks produce no duplicate records; a declined payment preserves the cart; a refund reverses the linked expense.	Gateway test log
Notifications	Every listed event delivers on enabled channels; preferences suppress correctly; duplicates are suppressed within five minutes; orphaned reminders are cancelled.	Test report
Document Management	Upload, encrypted storage, signed-URL retrieval, visibility control and expiry reminders all function; a non-member cannot enumerate or retrieve any document.	Test report, security suite
Travel Memories	Uploads resolve to the correct itinerary day in 100 percent of test cases; the album groups by day; deletion removes from storage within 24 hours.	Test report
Analytics and Administration	Account actions, moderation decisions and the audit log function; service health reflects all twelve services; personal spend dashboard renders from real trip data.	Demonstration, audit log export

6.2 Release Definition of Done

The release is complete only when every statement below holds. Each is binary; partial satisfaction counts as not done.

- Every Must-have requirement is implemented, tested and demonstrable.
- Every Should-have requirement is implemented or formally deferred with a recorded decision.
- Automated statement coverage on service business logic is at least 70 percent, and the full suite passes in continuous integration.
- All twelve services build, test and deploy independently, and each publishes an OpenAPI document.
- No critical or high severity vulnerability is present in application code or dependencies.
- All eight primary screens in Section 5 pass review at 360, 768, 1024 and 1440 pixels.
- An end-to-end scenario completes on the deployed environment: register, create a trip, invite, generate an itinerary, book, log expenses, settle and generate a journal.
- Every performance target in Section 2.1 is met under the stated load.
- Fault injection confirms that AI, partner and gateway failures degrade gracefully and create no partial records.
- No requirement is left without an owner, an implementing story and a verification method.
- User documentation and API documentation are published and current.
- The repository, Jira board and Google Classroom submission carry identical, current artefacts.

6.3 User Acceptance Test Scenarios

ID	Scenario	Pass condition
UAT-01	A group of five plans a seven-day trip end to end: create, invite, poll, generate, book, log, settle.	Completed without leaving Voyana and without facilitator intervention.
UAT-02	A first-time user creates a trip and generates an itinerary unaided.	Completed in under five minutes by at least 8 of 10 participants.
UAT-03	A group settles a trip carrying 30 mixed-currency expenses.	Every participant agrees the settlement figures; positions reconcile to zero.
UAT-04	A traveller adjusts a generated itinerary through the conversational assistant.	Requested change is applied only after explicit acceptance, and the cost consequence is stated.
UAT-05	The whole journey is completed on a 360 pixel mobile browser.	No horizontal scrolling, no truncated control, no blocked action.
UAT-06	A booking is attempted while the partner sandbox is disabled.	A clear failure state appears and no partial booking or expense record is created.
UAT-07	A non-member attempts to open a trip by direct link.	Access is refused and no trip data appears in the response body.
UAT-08	An administrator moderates three queued items.	All three decisions carry a reason and appear in the audit log with the correct actor.